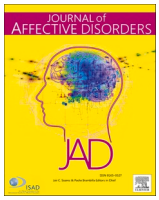
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Research paper



Depression in older adults: longitudinal study findings from a national survey in China

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ABSTRACT

Background: Late-life depression (LLD) is a significant global public health challenge among older adults. Identifying predictive/influential symptoms based on longitudinal data can contribute to early prevention, detection, and intervention of LLD. This study examined significant influential symptoms of LLD using cross-lagged panel network (CLPN) analysis based on a longitudinal national survey.

Methods: Data from the 2018 and 2020 waves of the China Health and Retirement Longitudinal Study (CHARLS) were used to construct a longitudinal depressive symptom network. Depressive symptoms were assessed using the 10-item Center for Epidemiological Studies Depression Scale (CESD-10). In expected influence (in-EI) and out expected influence (out-EI) were used as centrality indices to characterize interactions within the depressive symptom network. Network stability was assessed using the case-dropping bootstrap method.

Results: A total of 6533 older adults were assessed (Model 1). The most predictive/influential symptoms were "Everything was an effort" (CESD4), "Felt depressed" (CESD3) and "Felt lonely" (CESD9), all of which predicted increased risk for other depressive symptoms two years later. Additionally, when examining older adults with depression at baseline (Model 2), "Bothered by things" (CESD1), "Everything was an effort" (CESD4) and "Felt unhappy" (CESD8) emerged as the most predictive/influential symptoms of other depressive symptoms in the future.

Conclusions: Considering difficulties in treating LLD and its negative consequences on daily life in older adults, timely interventions targeting identified most influential symptoms may be conducive to the prevention and mitigation of future depression.

1. Introduction

Depression is one of the common mental health problems contributing to disability and mortality worldwide (Thornicroft et al., 2017). Late-life depression (LLD) refers to depression experienced by individuals aged 60 years and older (Wen et al., 2022). In recent years, in parallel with the rapidly aging global population, LLD has become a major clinical and public health concern (Taquet et al., 2020). Epidemiological data have found the global community prevalence of LLD to vary from 2.8 % to 35.0 %, with a pooled average prevalence of 13.5 %

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(Beekman et al., 1999). LLD is associated with a number of adverse outcomes such as increased risk of obesity, frailty, cognitive impairment (Alexopoulos, 2019; Buigues et al., 2015), and comorbid physical diseases, including cardiac and cerebrovascular disease, stroke, diabetes, Alzheimer's disease, and vascular dementia (Buigues et al., 2015; Daskalopoulou et al., 2016; Galts et al., 2019; Pan et al., 2011). Without timely intervention, LLD can seriously compromise physical health, quality of life, and functional independence in addition to increasing risk for suicide (Taylor, 2014).

Network analysis is a novel, advanced data analysis approach that has been widely used in the fields of psychiatry and psychology (Contreras et al., 2019). Network analyses examine psychiatric disorders or syndromes at the symptom level and explore inter-relationships among multiple symptoms within and between disorders (Beard et al., 2016). Such approaches can directly illustrate distinct symptoms (defined as nodes) and symptom-symptom interactions (defined as edges) using ordered spatial networks (Epskamp et al., 2017). Consequently, the most central symptoms or "core drivers" of psychiatric disorders/syndromes can be identified and prioritized in subsequent treatments, in order to reduce symptom severity and relapse risk (Borsboom and Cramer, 2013). To date, network analysis has been widely applied to LLD. For example, a study involving Korean older adults identified "Felt unhappy," "Hopelessness" and "Emptiness" as the most central LLD symptoms (Kim et al., 2021). Another study highlighted "Anhedonia," "Sad mood" and "Guilt" as the most central symptoms among Australian older adults with cancer (Sharpley et al., 2023).

However, traditional network analysis studies have notable limitations. In particular, an over-reliance on cross-sectional data limits the capacity to identify symptoms that predict or trigger other symptoms at later points in time (Epskamp et al., 2018b). For example, although there are cross-sectional relations between depressive symptoms such as sleep disturbances, loss of concentration, and anhedonia, previous research has suggested that initial sleep disturbances might lead to later losses of concentration and exacerbations in anhedonia (Borsboom and Cramer, 2013). Identifying and targeting "trigger" symptoms is potentially critical for preventing subsequent symptom activations and effectively managing the progression of disorders. Relatedly, traditional cross-sectional network analysis necessarily disregards directionality of symptom relationships. For instance, in an undirected network, if symptom A is closely linked to symptom B, it may mean that A leads to B or, conversely, that B leads to A. Although the symptoms are related, such data are insufficient for determining whether symptom A or B is the precursor to the later symptom. Furthermore, if unmeasured symptom C affects both A and B, interventions targeting symptom A may not be useful in preventing symptom B. Consequently, centrality index assessments in cross-sectional undirected network studies often provide limited insights for disorder prevention and intervention (Funkhouser et al., 2021).

In recent years, the cross-lagged panel network (CLPN) approach, based on directed network and longitudinal data, has addressed drawbacks of traditional cross-sectional network analyses. As a longitudinal network approach, CLPN is often applied to explore temporal predictive relationships and dynamic interactions between multiple variables in network models of mental health problems (Rhemtulla et al., 2017). The CLPN approach is based on Structural Equation Models (SEM) or Generalized Linear Models (GLM) and presents results though a directed weighted network graph. By establishing a node/variable regression model, CLPN approaches can estimate variable autoregression and cross-lagged effects by regressing data at a latter timepoint (T2) onto a previous timepoint (T1) (Funkhouser et al., 2021; Rhemtulla et al., 2017). This approach allows for probing predictive relationships among symptoms and assessing mutual predictive effects, potentially enhancing understanding of mental disorder development and maintenance (Williams et al., 2019). In CLPN network models, in expected influence (in-EI) and out expected influence (out-EI) serve as centrality indices to distinguish the "most influenced symptoms" and "most influential symptoms", respectively. Symptoms with highest out-EI are hypothesized to play vital roles in the onset and progression of disorders and syndromes, making them potential targets for clinical intervention (McNally, 2016).

Several studies have applied the CLPN approach to LLD. For example, one study conducted among older British adults found that "Everything was an effort", "Could not get going" and "Loneliness" showed strongest out-EI, indicating these were the most influential symptoms in the network model (Schlechter et al., 2023). Another study conducted among the older American adults showed that higher levels of "Web-based communication" predicted lower levels of depression in the future, underscoring the potential of web-based social connectedness as a means of combatting depression in this population (Hwang et al., 2021). Although these findings are informative, it is not clear whether they are applicable to non-Western countries where much of the planet's population is concentrated and cultural norms that often differ from those in the U.S. and U.K. (Cai et al., 2024). For instance, in Western societies, older adults generally exhibit higher acceptance of mental health services and greater willingness to actively seek professional help (Chow et al., 2019). In contrast, depression treatment-seeking rates remain substantially lower among Asian older adults (Morita et al., 2022). Furthermore, in China's healthcare system, primary care institutions (e.g., community health centres) typically prioritize chronic illness management while lacking the capacity for adequate mental health service provision (Chen et al., 2022). These cultural and structural differences between different societies underscore how depression markers, prevention efforts and intervention strategies developed for Western older adults are not necessarily applicable for older adults in non-Western countries such as China.

To the best of our knowledge, no studies to date have employed CLPN to explore LLD among older adults in highly populated understudied Asian countries including China. Based on this impetus, we explored the temporal and predictive relationships between depressive symptoms using longitudinal data from a national survey study of older Chinese adults to provide a novel perspective on LLD occurrence and progression in addition to generating empirically grounded hypotheses about prevention and interventions salient to a Chinese cultural context.

2. Materials and methods

2.1. Data source and study population

This study was a secondary analysis of data from the China Health and Retirement Longitudinal Study (CHARLS), a national survey conducted among community-dwelling older adults in China (Peking University Open Platform, 2024). The CHARLS was initiated in 2011; follow-up surveys were conducted in 2013, 2015, 2018 and 2020 (Hu et al., 2022). To ensure a representative sample, 28 out of 32 provinces were included using a multistage probability sampling method (Peking University Open Platform, 2024). Details of the study design, procedure and sampling have been reported previously (Zhao et al., 2014). The CHARLS research protocol was approved by the Biomedical Ethics Committee of Peking University (IRB00001052-11015). All participants provided written informed consent for participation.

In this study, data collected in the 2018 survey (Wave 4 of CHARLS) were designated as the "baseline assessment" and utilized with data gathered in 2020 (Wave 5 of CHARLS) to construct CLPN models. Participants were included if they 1) participated in both 2018 and 2020 surveys, 2) completed both depression assessments, and 3) were aged 60 years or older at the 2018 baseline. The sample selection flowchart is shown in Fig. 1.

2.2. Measurement tools

Severity of depressive symptoms was measured using the 10-item

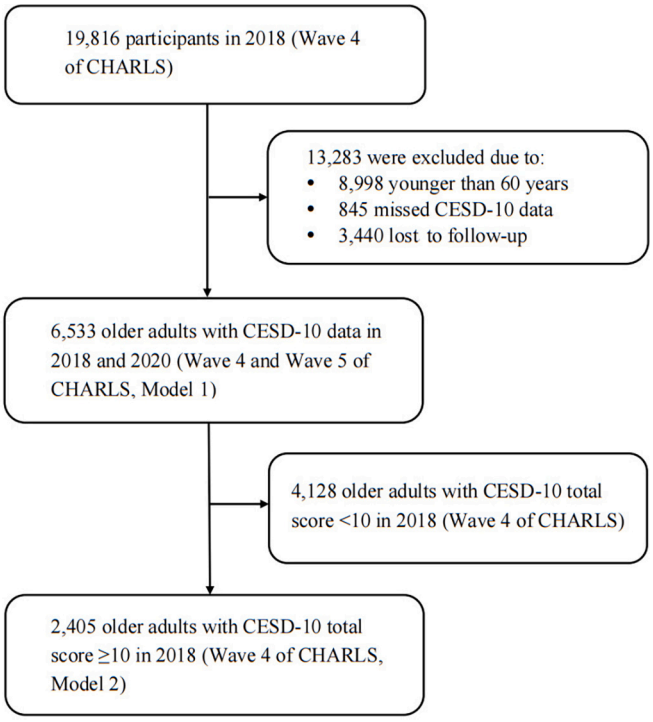


Fig. 1. The flowchart of this study.
Note: CHARLS: China Health and Retirement Longitudinal Study; CESD-10: 10-item Center for Epidemiological Studies Depression scale.

Chinese version of the Center for Epidemiological Studies Depression Scale (CESD-10) (Boey, 1999; Radloff, 1977). CESD-10 items include 1) Bothered by things; 2) Trouble concentrating; 3) Felt depressed; 4) Everything was an effort; 5) Hopeless about future; 6) Felt fearful; 7) Restless sleep; 8) Felt unhappy; 9) Felt lonely; 10) Could not get going. Items 5 and 8 of the CESD-10 were reverse-scored and rephrased from their original versions. CESD-10 items were scored from "0" (rarely or none of the time) to "3" (most or all of the time). Total CES-D scores range from 0 to 30 with higher total CESD-10 scores reflecting more severe depressive symptoms. Validation research conducted among Chinese older adults has found the CESD-10 to have satisfactory psychometric properties (e.g., Cronbach's alpha of $\alpha = 0.815$) A total score of ≥10 has been established as a cut-off for "having depression" in past published work (Boey, 1999; Huang et al., 2015).

2.3. Longitudinal network analysis

Longitudinal network analyses were performed using R software (version 4.3.3) (R Core Team, 2024). CLPN were generated to unravel connections between baseline and follow-up symptom assessments using the *glmnet* package (version 4.1-8) (Friedman et al., 2010). A CLPN can depict how a single node (i.e., symptom) at the initial time point predicts other nodes in the network model at a subsequent time point after adjusting for all other baseline measures (Niu et al., 2023). For each node, two centrality indices were computed: in-EI and out-EI. In-EI estimates the extent to which a symptom is predicted by other symptoms (i.e., the sum of values of incoming edges associated with one symptom) while out-EI provides an estimate denoting the extent to which a symptom can predict other symptoms (i.e., the sum of values of outgoing edges associated with one symptom) (Niu et al., 2023). These centrality estimates provide potential insights into the relative importance of each symptom from a network model and help to identify key nodes that play a critical role in temporal dynamics of symptom networks. Directed edges of each node pointing to itself signify autoregressive coefficients, which can be used to predict whether one symptom recurs in a later wave; in contrast, directed edges pointing to other nodes signify cross-lagged coefficients, which were used to predict the effect of one symptom on the occurrence of other symptoms in a subsequent wave. Arrow colors represent the directionality of effects, with blue arrows reflecting positive associations and red arrows indicating negative associations. Thicker edges reflect stronger correlations. To streamline the network models, an Extended Bayesian information criterion (EBIC) model, and a graphical least absolute shrinkage and selection operator (gLASSO) method (Epskamp and Fried, 2018; Garabiles et al., 2019), were applied to generate estimates. R package, *qgraph* (version 1.9.8), was used for network visualization (Epskamp et al., 2023).

All eligible older adults were included in the initial CLPN model (Model 1). Considering the potentially distinct network structure of depression in currently depressed older adults (Malgaroli et al., 2021), an additional CLPN model was generated specifically for older adults with depression at the baseline (Model 2) so that the progression and transformation of individual symptoms over time could be investigated within a subgroup for whom depression was already present.

2.4. Network stability

To assess the accuracy and stability of the observed network models, R package *bootnet* (version 1.4.3) (Epskamp, 2020) was used based on 1000 bootstraps. The centrality indexes of in-EI and out-EI were assessed using correlation stability coefficients (CS-coefficients); values >0.25 indicated moderate stability and values >0.5 indicated strong stability (Epskamp et al., 2018a). A bootstrapped difference test was also conducted to assess the robustness of nodes and edges. Differences were significant between two nodes or between two edges if zero was not included in the 95 % confidence interval (95 %CI) of 1000 bootstraps. Edge accuracy was estimated with bootstrapped 95 %CIs; narrower confidence intervals suggested a more reliable network.

3. Results

3.1. Study sample

In total, 6533 older adults from the baseline (2018 wave) assessment were included in analyses (Model 1). Demographic and clinical characteristics of the sample are shown in Table 1. The mean age of participants was 67.61 years [standard deviation (SD) = 5.74 years], 52.06 % ($n = 3401$) of respondents were male, 80.18 % ($n = 5238$) were married and 29.71 % ($n = 1941$) obtained secondary school or above education.

Table 1
Demographic and clinical characteristics of participants at the baseline ($N = 6533$).

Variables	N	%
Male	3401	52.06
Married	5238	80.18
Education		
Primary school and below	4592	70.29
Secondary school and above	1941	29.71
Urban residence[a]	1674	25.74
Having religious belief	674	10.32
Having two or more major physical diseases[a]	968	19.27
Currently smoking[a]	1207	31.84
Currently drinking[a]	1615	40.73
CESD-10 total ≥ 10	2405	36.81

Variables	Mean	SD
Age (years)	67.61	5.74
CESD-10 total	8.49	6.61

Note: CESD-10: 10-item Center for Epidemiological Studies Depression scale; SD: standard deviation.
[a] The frequencies are calculated based on available data.

In addition, the mean CESD-10 total score was 8.49 (SD = 6.61), and the prevalence of depression (CESD-10 total ≥ 10) was 36.81 % (n = 2405) in 2018. This depressed subgroup (N = 2405) constituted the subgroup analyzed in Model 2.

3.2. Longitudinal network analysis for general older adult sample

Fig. 2 shows the CLPN model for the total older adult sample (Model 1), depicting symptom interactions from the baseline time point to the follow-up. The strongest association was CESD3 → CESD1 ("Felt depressed" to "Bothered by things", edge weight = 0.0846), followed by CESD4 → CESD3 ("Everything was an effort" to "Felt depressed", edge weight = 0.0797) and CESD4 → CESD2 ("Everything was an effort" to "Trouble concentrating", edge weight = 0.0713). Fig. S1 presents Model 1 with autoregressive pathways, indicating effects of symptoms on themselves over time. Notably, CESD7 ("Restless sleep", edge weight = 0.3374) emerged as the most prominent node, with the presence of "Restless sleep" at baseline emerging as a strong predictor of restless sleep at a two-year follow-up among older Chinese adults.

For the centrality indices, CESD4 ("Everything was an effort", out-EI = 1.522), CESD3 ("Felt depressed", out-EI = 1.105) and CESD9 ("Felt lonely", out-EI = 0.653) exhibited the highest out-EI value, indicating their potential influence as triggers for other depressive symptoms in the future. Conversely, symptoms with highest in-EI values were identified as CESD1 ("Bothered by things", in-EI = 1.320), CESD2 ("Trouble concentrating", in-EI = 1.047) and CESD3 ("Felt depressed", in-EI = 1.011), suggesting increased susceptibility to the influence of depressive symptoms present at the baseline assessment (Table S1).

3.3. Longitudinal network analysis for older adults with depression

Fig. 3 illustrates the CLPN model for the older adult subgroup with current depression (Model 2). The strongest associations were CESD4 → CESD10 ("Everything was an effort" to "Could not get going", edge weight = 0.0796), followed by CESD1 → CESD3 ("Bothered by things" to "Felt depressed", edge weight = 0.0591) and CESD10 → CESD9 ("Could not get going" to "Felt lonely", edge weight = 0.0581). Fig. S2 shows the autoregressive pathway for Model 2. The symptom with most notable autoregression was CESD7 ("Restless sleep", edge weight = 0.2871), consistent with Model 1.

For the centrality indices, CESD1 ("Bothered by things", out-EI =1.178), CESD4 ("Everything was an effort", out-EI = 1.018) and CESD8 ("Felt unhappy", out-EI = 0.526) had the highest out-EI values, suggesting these symptoms exerted the strongest influence as triggers for subsequent depressive symptoms. Symptoms with the highest in-EI values included CESD10 ("Could not get going", in-EI = 1.603), CESD3 ("Felt depressed", in-EI = 0.989) and CESD1 ("Bothered by things", in-EI = 0.829); as such, these symptoms were potentially more susceptible to the influence of baseline depressive symptoms (Table S1).

3.4. Network stability

In the assessment of network stability, an excellent level of stability was observed for Model 1 (CS-coefficient = 0.750 for out-EI and in-EI), as 75.0 % of the sample could be dropped without significant changes to the network structure (Fig. S3). An acceptable level of stability was observed for Model 2 (CS-coefficient = 0.439 for out-EI and in-EI); 43.9 % of the sample could be dropped without significant changes to the network structure (Fig. S4). Bootstrapped 95 %CIs indicated that the edge weights were reliable and stable. The bootstrap difference test for edges indicated that most edge weight differences were statistically significant for Model 1 (Fig. S5) and Model 2 (Fig. S6). Finally, bootstrapped difference tests for node out-EI and in-EI had statistically significant differences from one another in individual comparisons, indicating that the primary results were reliable for Model 1 (Fig. S7) and Model 2 (Fig. S8).

4. Discussion

To our knowledge, this was the first study to adopt a longitudinal network analysis approach in the evaluation of changes in depression symptoms among community-dwelling older Chinese adults based on a large national survey. In the initial CLPN model based on adults with highly variable depression levels (Model 1), "Everything was an effort" emerged as the most predictive/influential symptom, followed by "Felt depressed" and "Felt lonely"; the presence of these symptoms at baseline was indicative of increased risk for other depressive symptoms two years later. Extending previous network analysis results for depression in older British adults (Schlechter et al., 2023), this study confirmed the predictive utility of "Everything was an effort" and "Felt lonely" in LLD within a Chinese cultural context. As such, these findings provide preliminary evidence for a degree of cross-cultural and geographical generalizability in symptom networks of LLD. Moreover, in the subset of older adults with baseline depression (Model 2), "Bothered by things", "Everything was an effort" and "Felt unhappy" were the most predictive/influential baseline symptoms for subsequent depressive symptoms at follow-up.

The CLPN network analysis revealed distinct temporal dynamics among depressive symptoms in Chinese older adults. The strongest association emerged from "Felt depressed" to "Bothered by things",



Fig. 2. Cross-lagged panel network of depressive symptoms among Chinese older adults (Model 1, N = 6533).
Note: Symptom centrality estimates for the networks using z-values. CESD-10: 10-item Center for Epidemiological Studies Depression scale.

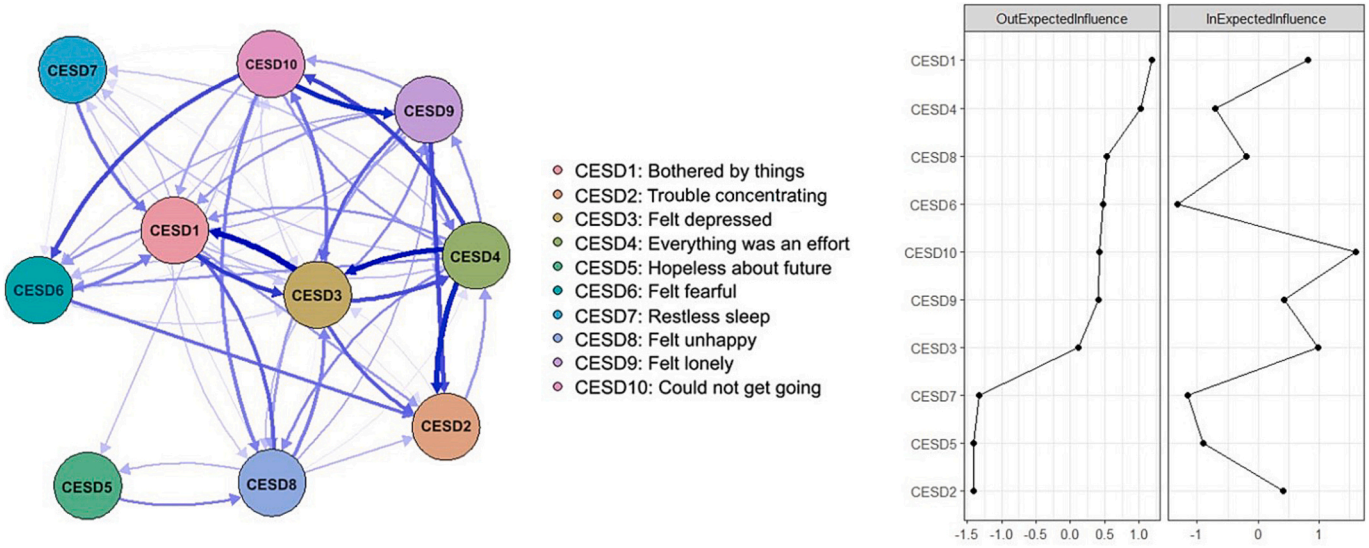


Fig. 3. Cross-lagged panel network of depressive symptoms among Chinese older adults with depression (Model 2, $N = 2405$).
Note: Symptom centrality estimates for the networks using z-values. CESD-10: 10-item Center for Epidemiological Studies Depression scale.

followed by "Everything was an effort" to "Felt depressed" and "Everything was an effort" to "Trouble concentrating". The "Felt depressed" → "Bothered by things" pathway could reflect a negative cognitive bias mechanism, whereby depressive moods could increase sensitivity to minor stressors, establishing a self-perpetuating vicious emotion-cognition cycle (Noworyta et al., 2021). The pathways from "Everything was an effort" to both "Felt depressed" and "Trouble concentrating" may have reflected an exacerbation in motivational-physical symptoms that precipitated subsequent deterioration in both affective and cognitive domains of depression (Schlechter et al., 2023). These interconnected pathways could potentially represent core mechanisms underlying symptom maintenance and exacerbation in Chinese older adults, highlighting the possible utility of future intervention research targeting these specific symptom transitions.

In both Model 1 and Model 2, "Everything was an effort" emerged as the most predictive/influential symptom, significantly linked to the future experiences of "Felt depressed", "Trouble concentrating" and "Could not get going". Perceiving everything as an effort, may reflect feelings of being overwhelmed, exhausted or bored with activities and has been termed as a "lack of energy" in previous research (Yang et al., 2024). Previous studies have also found that endorsements of "Everything was an effort" can serve as an indicator of persistent depression (Frank et al., 2021). In younger adults with depression, reports that "Everything was an effort" were a common complaint and the main reason for seeking medical assistance because these experiences interfere with work and life routines (Wielopolski et al., 2015). However, identifying this as a depressive symptom in older adults poses interpretive challenges because overlapping age-related declines and physical illnesses (Chen et al., 2024a, 2024b; Ehrenkranz et al., 2021; Zhang et al., 2024) may also contribute to perceptions that everything is an effort. For older individuals, viewing everything as an effort might limit motivation and engagement in activities or social interactions, impacting autonomy and independence (Anic et al., 2022), thus exacerbating feelings of frustration, helplessness, and loss of control (Scheetz et al., 2012). Conversely, the restoration of energy is a potential indicator of improvements in depression (Lam et al., 2017).

"Felt depressed" emerged as a predictive/influential symptom in Model 1, suggesting its pivotal role in subsequently triggering other depressive symptoms among older adults. Based on the direction of this arrow, baseline "Felt depressed" responses were related to future endorsements of "Bothered by things", "Everything was an effort" and "Trouble concentrating". This finding is consistent with several cross-sectional network analyses; for instance, a study based on the Health and Retirement Study (HRS) showed that "Felt depressed" was a central symptom of depression both before and after diagnosing diabetes, heart disease, stroke, and cancer among American older adults (Airaksinen et al., 2020). In another study involving Chinese older adults, "Felt depressed" was also the most central symptom of depression (Liu et al., 2024). According to diagnostic criteria for major depressive disorder (APA, 2024; WHO, 2024), persistent feelings of depression are a necessary and signature symptom for diagnosis. For older adults, feeling depressed can involve physical and mental domains, potentially influencing thinking and cognitive processes, fostering negative emotions, and exacerbating other depressive symptoms (Shao et al., 2022; Tanaka, 2018). Second, feeling depressed can also impede emotional expression and social interactions, leading to interpersonal isolation and loneliness, thereby worsening other depressive symptoms (Santini et al., 2020). Third, feeling depressed has links with impairments in appetite, sleep quality, and physical strength, deficits that compromise quality of life and increase risk for other depressive symptoms (Aprahamian et al., 2021; Bao et al., 2017; Veronese et al., 2017). As such targeting feelings of depression in early interventions for older adults may be crucial for preventing the amplification of other depressive symptoms.

"Felt lonely" emerged as another highly predictive/influential symptom in Model 1, particularly in relation to endorsing future experiences of "Trouble concentrating", "Felt depressed" and "Everything was an effort". "Felt lonely" also refers to loneliness (Lee et al., 2023) and has gained attention in psychogeriatrics due to its multifaceted impact on depression (Saul and Gursul, 2021). In the short term, loneliness contributes to increases in emotional distress, disappointment and desperation among older adults (Harris, 2024); chronic loneliness potentially contributes to various psychological and cognitive problems such as memory deficits, inattention and low self-worth that can exacerbate depressive symptoms (Stokes et al., 2023). Moreover, persistent loneliness has links with increased risk for psychosomatic symptoms (Pai and Vella, 2022). Because loneliness often stems from inadequate social interactions and support systems (Ohry, 2018), self-help efforts may be arduous and various depressive symptoms may be exacerbated over time if the desire for increased social contact is not addressed (Saul and Gursul, 2021).

"Bothered by things" was identified as a predictive/influential symptom among older adults with depression in Model 2, aligning with another CLPN analysis on depressed middle school students in China (Tao et al., 2023). In this study, baseline reports of being "Bothered by

things" were associated primarily with future endorsements of "Felt depressed", "Could not get going" and "Trouble concentrating" among depressed older adults. Being bothered by things reflects persistent perceptions of stress or ongoing tension in response to external changes (Carey, 2018; Kelly et al., 2021). Presumably, if continual stress states are not addressed, susceptibility to a range of depressive symptoms is increased among already depressed older adults (Cristóbal-Narváez et al., 2022). Being bothered by things can also disrupt emotion regulation and coping (Shaffer et al., 2022) in addition to increasing irritability and restlessness (Mishra and Varma, 2023), depleting emotional resilience and exacerbating other depressive symptoms (Martínez-Líbano and Yeomans, 2023).

"Felt unhappy", another key baseline predictive/influential symptom among older adults with depression (Model 2), was closely connected with follow-up reports of "Felt depressed", "Could not get going" and "Felt lonely". Accumulated feelings of unhappiness may perpetuate negative attitudes towards life among currently depressed people, possibly draining energy and culminating in depression (Brummer et al., 2014; Gould et al., 2018). This phenomenon is particularly pronounced among individuals who lack adaptive coping resources and emotional regulation capacities such as older adults with limited financial means, insufficient family support, or concurrent chronic physical illnesses (Cohen-Mekelburg et al., 2024; Giebel et al., 2022; Harris, 2024). Ongoing unhappiness can also impact physical well-being among older adults (Hsu et al., 2021). Previous studies have documented direct effects of unhappiness on physical health, including but not limited to weakened immune system function, dyspepsia, risk of cardiovascular and cerebrovascular events, enhanced pain sensitivity, and sleep disturbances (Lotfaliany et al., 2018; Smith et al., 2022; Stieglitz et al., 2015). These physical changes contribute to further stress response elevations and negative emotions, robbing older adults of their enthusiasm and vitality while potentially increasing their risk for depression (Smith et al., 2022).

To summarize, based on out-EI analyses, empirically based candidates for the most influential depression symptoms were identified, potentially serving as specific targets in future prevention and intervention studies of LLD. In clinical practice, targeted interventions should be implemented to address these symptoms. Specifically, for "Felt depressed", clinicians should regularly monitor affective changes and consider evidence-based psychotherapies (e.g., cognitive behavioral therapy) where necessary (Ji et al., 2023). For experiences of "Everything as an effort", behavioral activation therapy with structured goal-setting as well as the pursuit of personally identified pleasant activities could be utilized to enhance behavioral engagement (Ji et al., 2023). With respect to feelings of loneliness, community-based interventions including regular group activities and online social activities should be developed to strengthen social contact (Xu and Koszycki, 2023).

In addition, in-EI analyses revealed "Restless sleep" as having the highest autoregression across CLPN Model 1 and Model 2. This finding suggests that once older adults develop sleep disturbances, such problems persist for up to two years later. Sleep hygiene interventions may have utility in addressing chronic sleep disturbances. Finally, the identification of most influenced symptoms in Model 2 (i.e., "Could not get going", "Felt depressed" and "Bothered by things") suggests that depression relapses and exacerbations can be predicted by the emergence and/or aggravation of these symptoms. As such, effective management of these symptoms among older adults with recurrent depression is a potentially useful directive for future intervention research.

Strengths of this study included its large, nationally representative sample and the employment of CLPN models based on longitudinal data to illuminate specific depression symptoms most critical to increased risk for subsequent depressive symptoms two years later. However, several limitations should also be noted. First, the generalizability of findings might be limited to Chinese community-dwelling older adults and may not extend to hospitalized older adults or older samples in other countries. Second, the two-year interval between assessments might have masked certain short-term associations between symptoms because symptom fluctuation levels within this timespan are unknown. Third, self-report responses based on the CESD-10 were used to assess depression; hence, certain response biases (e.g., social desirability bias, recall bias) might affect the results. Fourth, the impact of potential confounding factors affecting symptom network models (e.g., physical comorbidities, cognitive impairments, socioeconomic status) were not assessed and/or could not be addressed because the current CLPN approach does not support subgroup analyses. Fifth, CLPN model results underscored group-level results rather than changes in particular individuals because this approach cannot disaggregate stable individual differences (between-person effects) from within-person effects (Wysocki et al., 2022). Future resolutions to this issue might necessitate the formulation of more sophisticated mathematical models for network analysis (Epskamp et al., 2018a).

In conclusion, "Everything was an effort", "Felt depressed" and "Felt lonely" were endorsed as the most predictive/influential baseline symptoms of other depressive symptoms at a two-year follow-up among Chinese older adults. In addition, for older adults with depression, "Bothered by things", "Everything was an effort" and "Felt unhappy" served as significant baseline indicators of potential future onsets of or exacerbations in additional depressive symptoms. Findings provide evidence-based foundations for testing effects of timely interventions targeting these symptoms in future research extensions on depression prevention and management in older adults.

CRediT authorship contribution statement

Meng-Yi Chen: Formal analysis, Writing – original draft, Conceptualization. **He-Li Sun:** Data curation. **Qinge Zhang:** Methodology. **Sha Sha:** Methodology. **Zhaohui Su:** Data curation. **Teris Cheung:** Data curation. **Gabor S. Ungvari:** Data curation. **Todd Jackson:** Writing – review & editing. **Yuan Feng:** Conceptualization. **Yu-Tao Xiang:** Conceptualization, Writing – original draft.

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Declaration of competing interest

The authors have no conflicts of interest to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.120082.

Data availability

All data supporting the findings of this study was available in the Peking University Open Platform: China Health and Retirement Longitudinal Study (CHARLS) (http://charls.pku.edu.cn/en.)

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